Title Slide

Project Review 1 • Online Healthcare Management System

Online Healthcare Management System (CareFirst)

Presenter: Deepanshu (Lead Developer & System Architect)

Course: Java Web-Based Application Development | Evaluation: Review 1

Stack: Jakarta Servlet 6.0 + JSP 3.1 + JDBC + HikariCP + MySQL 8.0

Rubric Mappings: Problem Understanding (8) • Core Java (10) • Database/JDBC (8) • Servlets (7)

Problem Statement & Objectives

Clinical Challenge & Strategic Engineering Objectives

- The Problem: Fragmented clinic scheduling, double bookings, missing medical records, lack of RBAC.
- The Solution: 100% Java web architecture with Jakarta Servlets, normalized 3NF schema, ACID slot locks.
- Architecture: Strict MVC separation (model -> dao -> service -> servlet) + filters and listeners.
- Connectivity: Leased HikariCP connection pool with try-with-resources and 100% PreparedStatement security.

User Roles & Key Features

Multi-Stakeholder Feature Matrix (Admin, Doctor, Patient)

- ■ Administrator: User and doctor management, analytics KPIs, audit logs, system settings, DB diagnostics.
- ■ Doctor: Weekly recurring availability scheduler, patient appointment queue, EHR records & prescriptions.
- ■ Patient: Specialist doctor catalog, live slot booking with concurrency lock, prescription and history review.

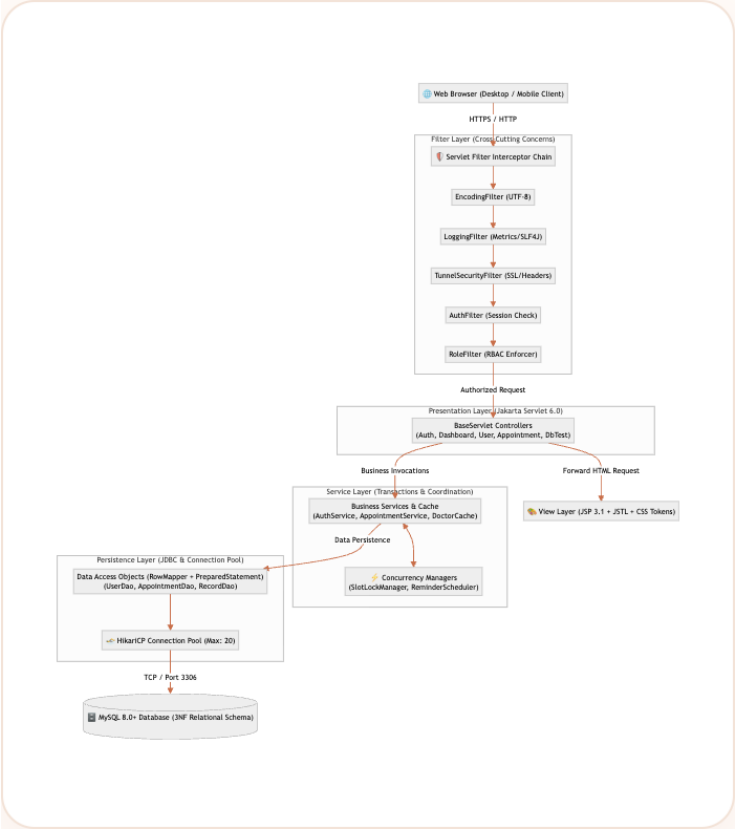
Tech Stack & Architecture Rationale

Enterprise Stack (Java 17, Servlet 6, JSP, JDBC, MySQL)

- Java 17 LTS: Modern OOP, type safety, multithreading, and high performance.
- Jakarta Servlet 6.0: Fine-grained HTTP request lifecycle control and filter interceptor chains.
- Jakarta JSP 3.1 & JSTL 3.0: Dynamic server-side templating with automatic XSS output escaping.
- JDBC & HikariCP 5.1: Zero ORM overhead, connection pooling, PreparedStatement query parameters.
- MySQL 8.0+ / 9.x: Relational ACID storage engine with InnoDB foreign keys and B-Tree indexes.

System Architecture (MVC Layered)

Layered MVC Architecture & Security Interceptor Pipeline



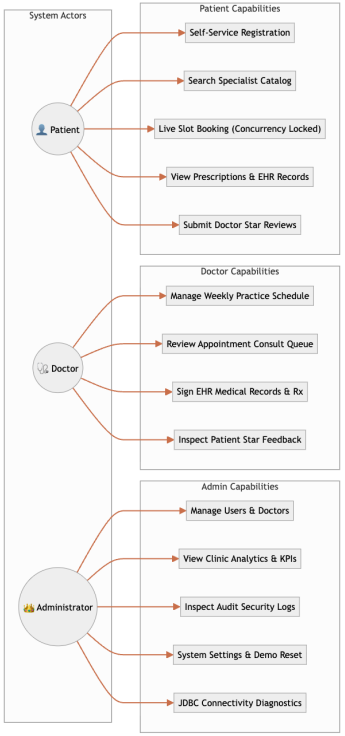
Project Structure & Packages

Clean Modular Project Structure & Layer Packaging

- model: Polymorphic user hierarchy (Doctor, Patient, Admin) and interfaces (Bookable, Reportable).
- dao: Generic Repository<T, ID> and functional RowMapper<T> for type-safe JDBC extraction.
- service: Business invariant validation, caching (DoctorCache), and multi-table transactions.
- servlet: HTTP presentation controllers inheriting lifecycle utilities from BaseServlet.
- filter: UTF-8 encoding, SLF4J request logging, tunnel security, auth check, and RBAC guard.
- listener: AppContextListener for startup/shutdown; AppSessionListener for active session tracking.
- util: HikariCP DBConnection, PasswordUtil (BCrypt), CsrfUtil, ApiResponse, OpenPDF generator.

System Use-Case Modeling

Role-Based Access Control Actor Interactions



Database Design: ER Diagram

3NF Normalized Relational Schema Design

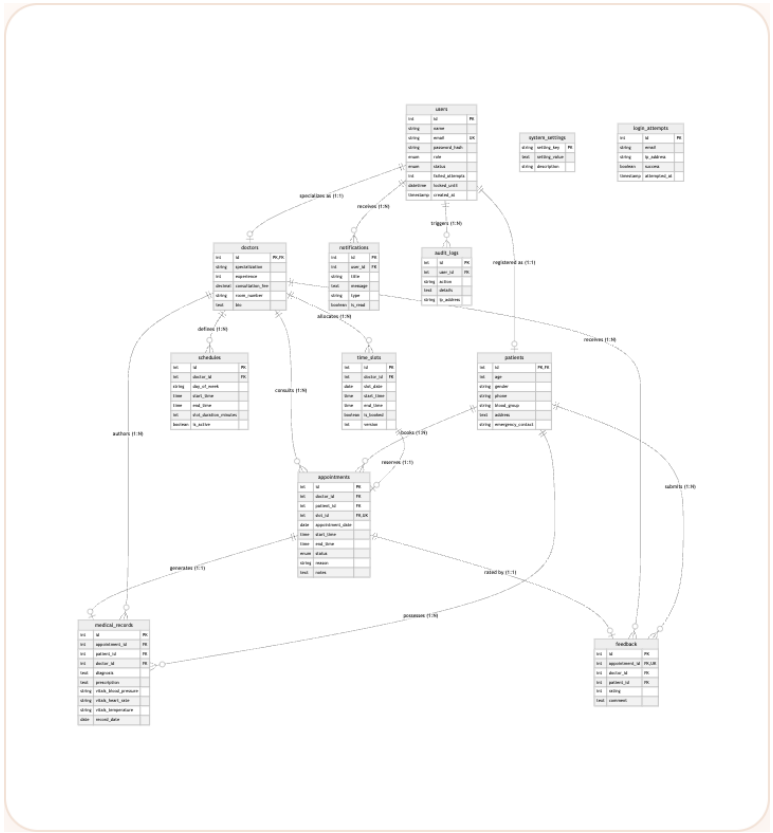


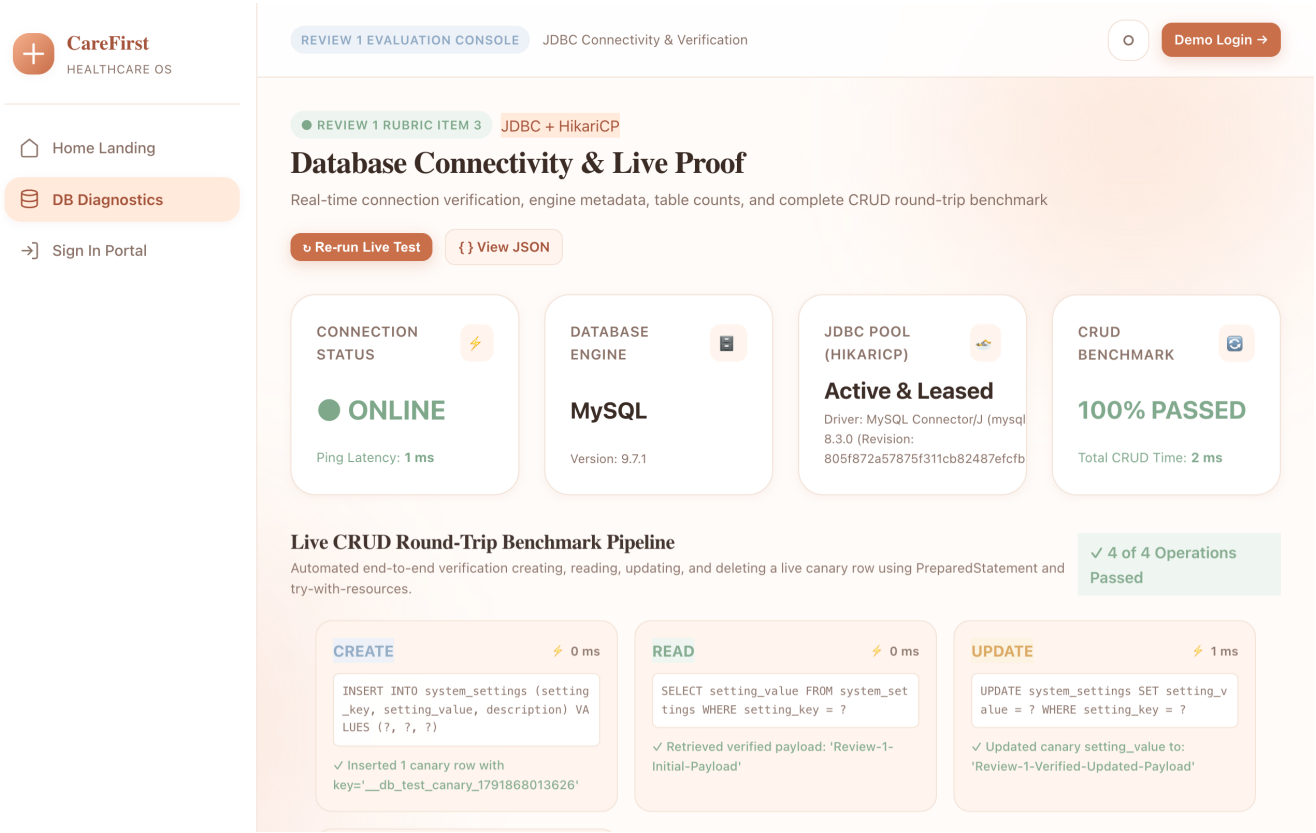
Table Overview & Constraints

12-Table Schema Specifications & Foreign Keys

- users (16 rows) | doctors (5 rows) | patients (10 rows) | schedules (21 rows)
- time_slots (30 rows) | appointments (11 rows) | medical_records (5 rows) | feedback (5 rows)
- notifications (190 rows) | system_settings (11 rows) | audit_logs (8 rows) | login_attempts (0 rows)
- Constraints: PKs on all tables, 15 Foreign Keys with cascading actions, UNIQUE doctor slot locks, CHECK ratings.

Database Connectivity (JDBC)

HikariCP Connection Pool & /db-test Diagnostic Proof



Core Java Concepts

OOP, Generics, Exceptions, Multithreading & Transactions

- OOP & Polymorphism: User base class specialized by Doctor, Patient, and Admin with contract interfaces.
- Generics & Collections: Generic Repository<T, ID>, RowMapper<T>, ApiResponse<T>, ConcurrentHashMap.
- Custom Exception Hierarchy: HealthcareException checked root with domain subclasses for auth, validation, and conflicts.
- Multithreading & Synchronization: SlotLockManager ReentrantLocks preventing concurrent double-booking clashing.
- Transaction Management: Explicit conn.setAutoCommit(false), commit(), and rollback() in multi-step booking operations.

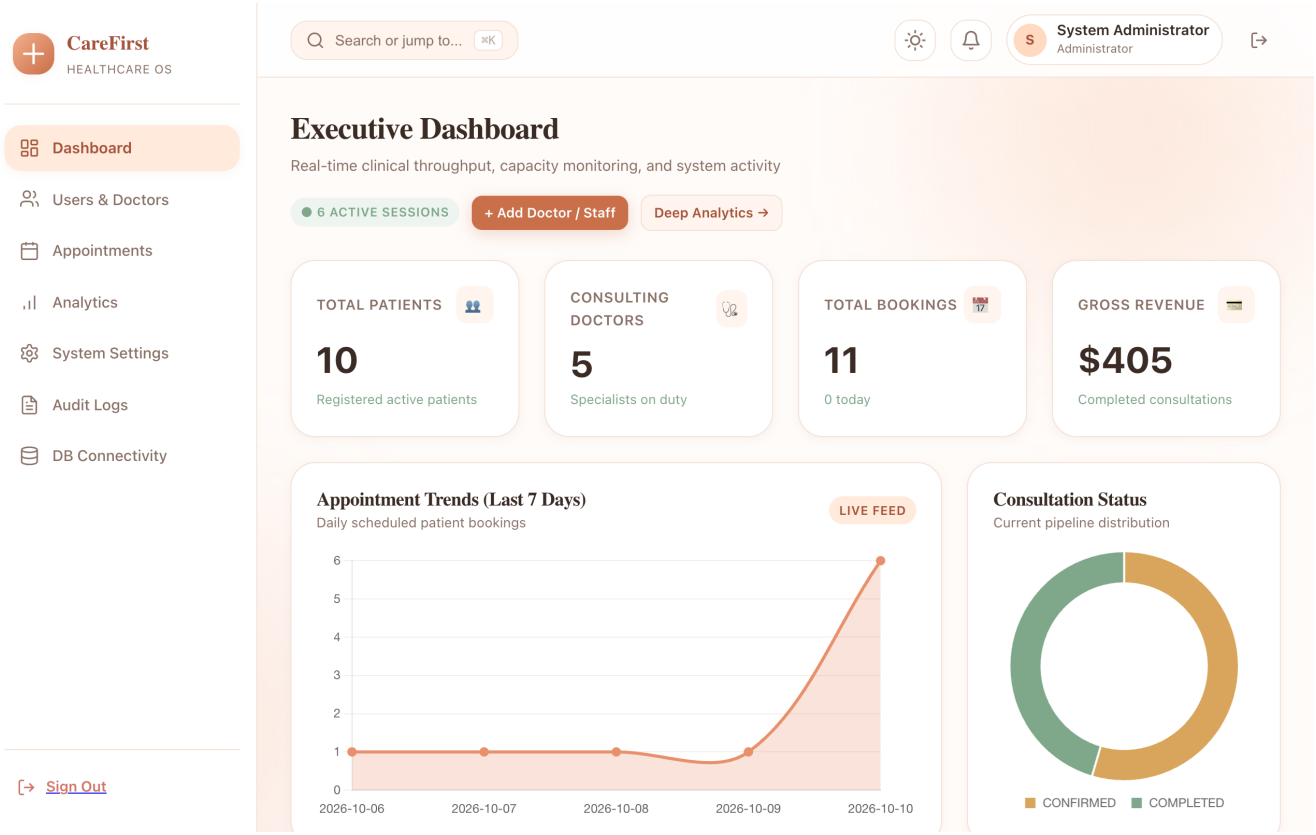
UI/UX Design System

Warm Peach Design Tokens, Bento Grid & Typography

- Warm Peach Aesthetic: Custom CSS design tokens (--bg: #FFFBF8, --peach-500: #E8906B, --text: #3A2B25).
- Modern Typography: Fraunces serif headings paired with Plus Jakarta Sans body typography.
- Bento Grid Layout: Modular card spans with animated KPI counters and activity streams.
- Accessibility & Themes: High contrast compliance, micro-animations, and full dark mode support.

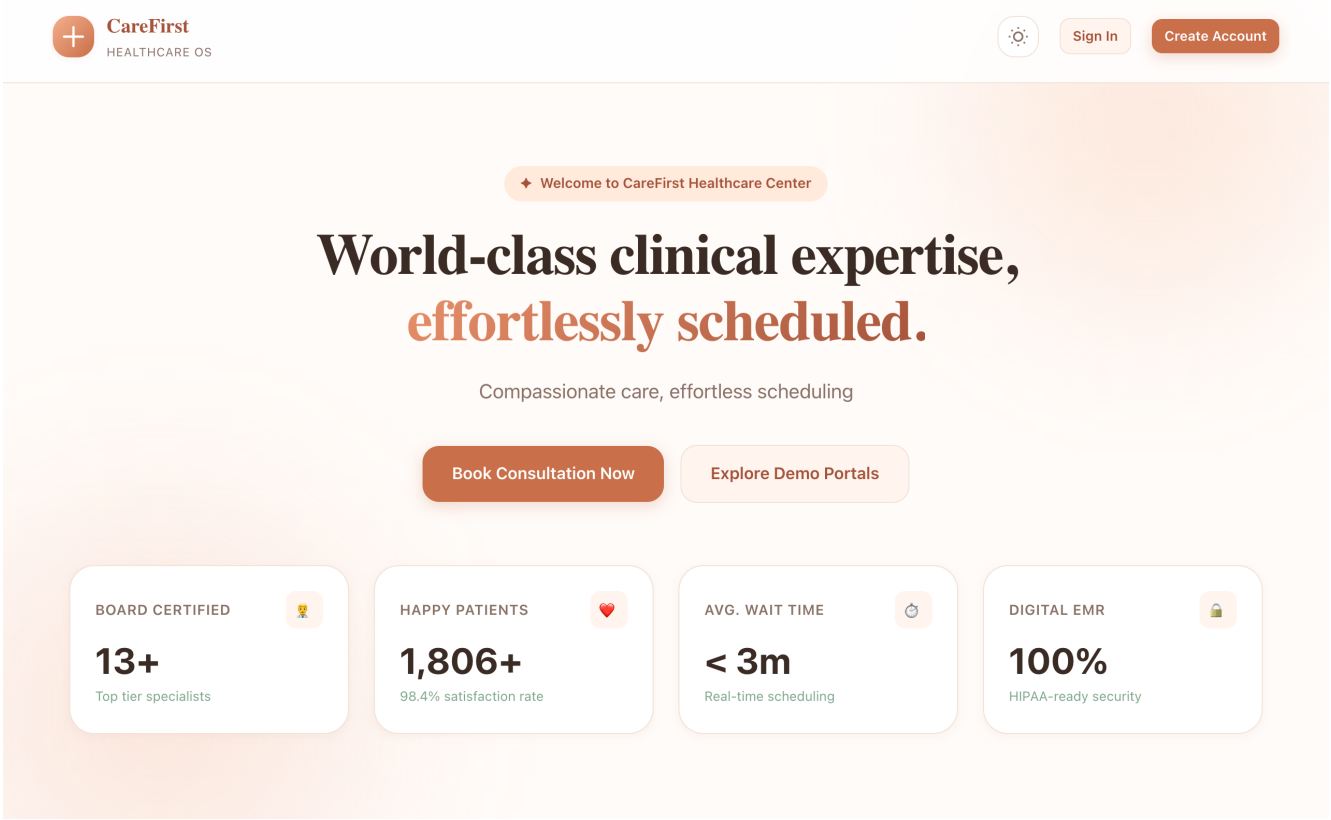
Desktop Application Walkthrough

Multi-Role Dashboards Desktop Experience



Responsiveness & Mobile Adaptation

Mobile (375px) vs Desktop (1280px) Responsiveness



Servlets & Web Integration

Lifecycle, Session Management, CSRF & Filters

- Servlet Lifecycle: Explicit overrides of init(), service(), doGet(), doPost(), and destroy() with SLF4J logging.
- Session Management: 30-minute HTTP sessions with session fixation defense (changeSessionId()).
- Cookie Hardening: HttpOnly, SameSite=Lax, and dynamic Secure attributes over HTTPS tunnels.
- Filter Pipeline: Chain of responsibility (EncodingFilter -> LoggingFilter -> TunnelFilter -> AuthFilter -> RoleFilter).

Milestones & Review 2 Plan

Current Accomplishments & Review 2 Development Roadmap

- Review 1 Status: 100% of architectural, database, connectivity, and UI/UX requirements completed and verified.
- Review 2 Focus: WebRTC telehealth video calls, automated email/SMS reminders, Chart.js analytics, OpenPDF reports.

Repository & Demo Credentials

Public GitHub Repository, Verification & Demo Access



Repository: https://github.com/ParvPrakash/healthcare-management-system-

Admin: admin@carefirst.com / Admin@123

Doctor: doctor.sarah@carefirst.com / Doctor@123

Patient: john.doe@example.com / Patient@123

One-Command Local Run: mvn compile exec:java